

FSO COMMUNICATION SYSTEM

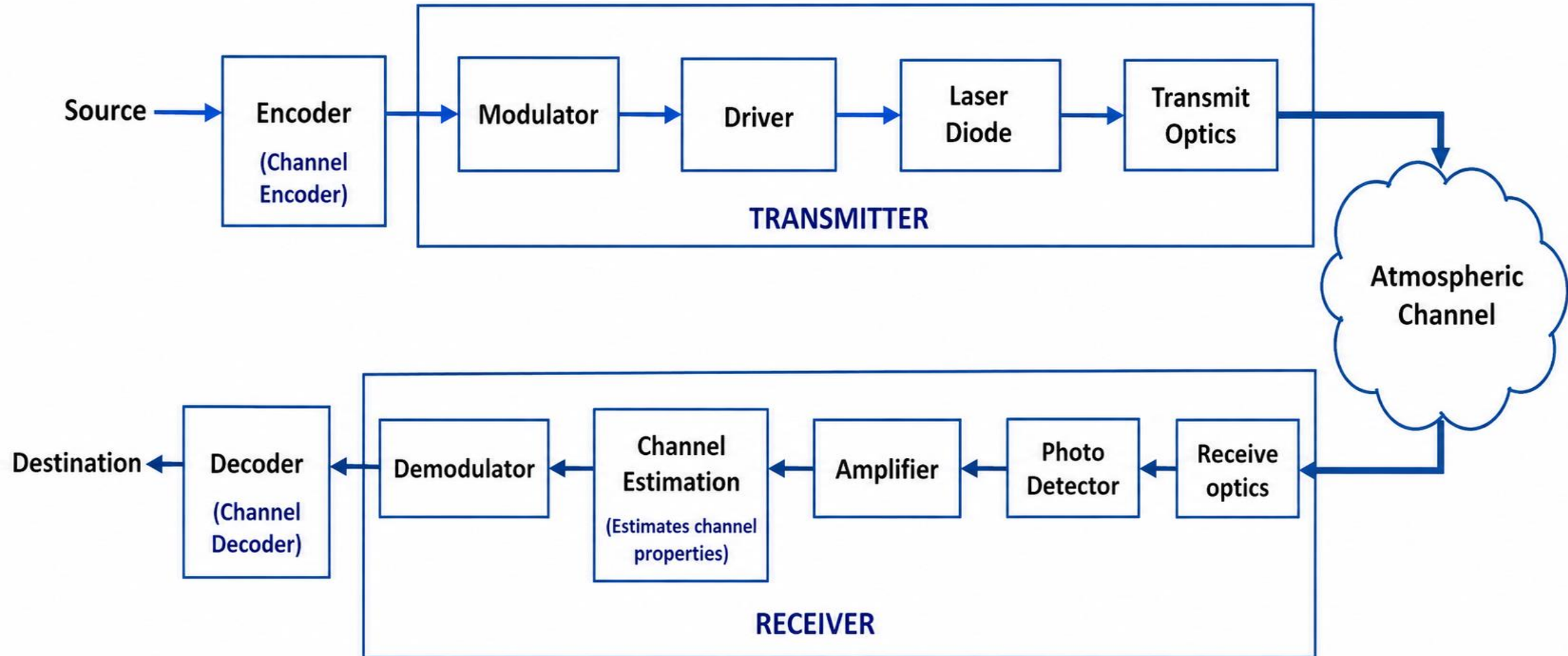
ABOUT FSO

- A wireless technology that uses light beams from lasers or LEDs to transmit data through the air, outer space or a vacuum.
- Requires a straight, clear path between the sender and receiver.
- Heavy fog, rain, snow or dust can scatter the light beam and weaken the signal

COMMON USES:

- Space and Satellites
- Urban Networking
- Temporary or Emergency Links

BLOCK DIAGRAM



SOURCE

- Produces the original information.
- Information may be text, voice, image or video.
- Sends the data for processing.

CHANNEL ENCODER

- Adds redundant bits using mathematical calculations.
- Improves communication reliability.
- Helps detect and correct transmission errors.
- Protects data against atmospheric impairments.

Examples: Low Density Parity Check (LDPC), Reed–Solomon, Turbo, Polar.

MODULATOR

- Receives encoded bits.
- Maps bits into optical pulses.
- Makes the data suitable for laser transmission.
- Common methods: On-Off Key, Pulse Position Modulation, Orthogonal Frequency-Division Multiplexing.

DRIVER

- Provides the required driving current.
- Converts modulation signals into laser drive signals.
- Ensures stable laser operation.
- Protects the laser from excessive current.

LASER DIODE

- Converts electrical signals into laser light.
- Produces a narrow optical beam.
- Supports high-speed communication.
- Common wavelengths: 850 nm and 1550 nm.

TRANSMIT OPTICS

- Focuses the laser beam.
- Reduces beam divergence.
- Improves transmission efficiency.
- Directs the beam toward the receiver.

ATMOSPHERIC CHANNEL

- Laser travels through free space.
- Introduces attenuation.
- Causes scattering and absorption.
- Produces atmospheric turbulence.
- Affected by fog, rain, dust and smoke.
- May introduce pointing errors.

ATMOSPHERIC TURBULENCE

Atmospheric turbulence is the irregular movement of air that makes the laser beam unstable while travelling from the transmitter to the receiver.

RECEIVE OPTICS

- Collects the transmitted optical signal.
- Focuses light onto the photodetector.
- Increases received optical power.
- Improves signal collection.

PHOTO DETECTOR

- Receives the laser beam.
- Converts light into electrical current.
- Produces the received electrical signal.
- Common devices: Photodiodes

AMPLIFIER

- Amplifies weak signals.
- Improves signal level.
- Increases Signal-to-Noise Ratio (SNR).
- Prepares the signal for processing.

CHANNEL ESTIMATION

- Uses known pilot symbols.
- Estimates channel gain.
- Measures attenuation.
- Estimates noise level.
- Evaluates turbulence effects.
- Does not recover the transmitted data.

AVAILABLE TECHNIQUES IN CHANNEL ESTIMATION

1. Least Squares (LS)
2. Minimum Mean Square Error (MMSE)
3. Kalman Filter
4. Maximum Likelihood (ML)
5. Blind Channel Estimation
6. Semi-Blind Estimation
7. Machine Learning-based Estimation
8. Deep Learning-based Estimation

BASIC WORKFLOW

- Channel estimation does not estimate the properties of the transmitted signal itself. The transmitted signal is already defined by the transmitter.
- It estimates how the communication channel has modified that transmitted signal.
- For example:
- Channel gain: How much the signal amplitude changed.
- Attenuation: How much power was lost during propagation.
- Noise level: How much unwanted electrical/optical noise was added.
- Turbulence effects: How strongly atmospheric turbulence distorted the signal.
- These estimates allow the receiver to compensate for channel impairments before demodulation and decoding.

EQUALIZER

- Uses channel estimation results.
- Compensates channel attenuation.
- Reduces distortion.
- Improves signal quality.
- Produces a cleaner signal.

EQUALIZER

- The equalizer does not recover the original data.

Example: Original data = 1011

- After encoding and transmission through the atmosphere, the received signal is distorted.
- The equalizer tries to make that distorted signal closer to what was originally transmitted.
- But it does not say: "The original data was 1011"

DEMODULATOR

- Detects transmitted symbols.
- Converts optical symbols into bits.
- Makes bit decisions.
- Produces the received bit stream.

CHANNEL DECODER

- Uses redundant bits added by the encoder.
- Detects bit errors.
- Corrects transmission errors.
- Recovers the original information bits.
- Improves communication reliability.

DESTINATION

- Displays the received data.
- Final user receives the information.
- Communication process ends.

RF & FSO

RF (Radio Frequency) communication means sending information using radio waves through the air. Technologies such as 4G, 5G, Wi-Fi, Bluetooth and radio communication use RF. An RF transmitter converts data into a radio signal, sends it through an antenna and the receiver captures and processes that signal.

There is no need to completely replace RF with FSO. Instead, FSO can be used where it has advantages over RF. FSO uses a laser beam, which can provide very high data rates, large bandwidth, highly directional and secure communication and very low electromagnetic interference. However, FSO is strongly affected by fog and atmospheric conditions and requires accurate line-of-sight alignment. Therefore, FSO and RF are better viewed as complementary technologies and hybrid RF–FSO systems can use FSO when conditions are good and RF as a backup when the optical link is degraded.

COMPARISON

Channel Estimation	Channel Classification
Finds the numerical properties of the channel.	Finds the category/class of the channel.
Estimates how much the signal was affected.	Identifies what condition the channel is in.
Output is usually numerical values.	Output is usually a class/label.
Used for channel compensation/equalization.	Used for choosing an appropriate communication strategy.
Example: Channel gain = 0.65	Example: Channel = Moderate turbulence

WORKFLOW EXAMPLE

- **1. Source — “HELLO”**

The user types the message **HELLO**.

- **2. Convert to bits**

The computer represents the message in binary.

- **3. Channel Encoding**

Extra redundant bits are added to protect the data from errors.

- **4. Modulation**

The encoded bits are converted into optical symbols, e.g., laser ON/OFF pulses.

- **5. Laser Transmission**

The laser diode produces the corresponding light pulses and sends them through the transmit optics.

- **6. Atmospheric Channel**

The laser travels from Building A to Building B. Fog, turbulence, rain, etc. may weaken or distort the signal.

- **7. Photodetector**

The receiver collects the laser and converts it into an electrical signal.

- **8. Amplifier**

The weak electrical signal is amplified.

- **9. Channel Estimation**

Using known pilot signals, the receiver estimates how the atmosphere affected the signal for example, channel gain, attenuation and noise.

- **10. Equalization**

The estimated channel information is used to compensate for the channel effects.

- **11. Demodulation**

The compensated signal is converted back into received bits.

- **12. Channel Decoding**

The decoder uses the redundant bits to detect/correct errors.

- **13. Source Decoding**

The recovered bits are converted back into the original message.

- **14. Destination**

The user at Building B sees: HELLO

THANK YOU